a) Let X reflect the matrix of $N=3$ non-redundant asset payoffs ($n \times j$), we have:

$$X = \begin{pmatrix} 10 & 10 & 0 \\ 20 & 10 & 10 \\ 30 & 10 & 10 \end{pmatrix}$$

We only have complete market if and if only X has full rank and is invertible. Indeed, its determinant is:

$$\begin{aligned} \det(X) &= 10 \begin{vmatrix} 10 & 10 \\ 10 & 10 \end{vmatrix} - 10 \begin{vmatrix} 20 & 10 \\ 30 & 10 \end{vmatrix} + 0 \\ &= 0 - 10(200 - 300) + 0 \\ &= 1000 \end{aligned}$$

Since its determinant is $\neq 0$, the market is complete.

b) Since Law of one price holds in this economy, we have:

$$\begin{aligned} P_C &= X^{-1} \cdot P \\ &= \begin{pmatrix} 0 & -0.1 & 0.1 \\ 0.1 & 0.1 & -0.1 \\ -0.1 & 0.2 & -0.1 \end{pmatrix} \begin{pmatrix} 16 \\ 8 \\ 8 \end{pmatrix} = \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} \end{aligned}$$

c) Since LOP holds, price of asset D in this economy is:

$$P = X' P_C = \begin{pmatrix} 30 & 20 & 10 \end{pmatrix} \begin{pmatrix} 0.2 \\ 0.4 \\ 0.2 \end{pmatrix} = 16$$

d) We have $P = \sum_{n=1}^{N} P_C(n) \cdot X(n) = \sum_{n=1}^{N} \pi(n) \cdot \underbrace{\frac{P_C(n)}{\pi(n)}}_{M(n)} \cdot X(n)$

As a result, the realization of M under the three states of nature is:

$$M_1 = \frac{0.2}{1/4} = 0.8$$

$$M_2 = \frac{0.4}{1/2} = 0.8$$

$$M_3 = \frac{0.2}{1/4} = 0.8$$

This economy satisfies No Arbitrage since we observe that an asset with Payoff ≥ 0 and a chance for pay off >0 has a price >0
Indeed, the contingent claim prices $P_c(n)$ are all strictly positive, thus guaranteeing strictly positive asset price.
The SDF is the same across all states, which is consistent with the existence of a unique and positive SDF under complete markets and no-arbitrage.

e) Based on the SDF equation $P_t = E_t[M_{t+1} \cdot X_{t+1}]$, the price of asset D is:

$$P_D = E_t(M \cdot X_D)$$

$$= \frac{1}{4}(0.8)(30) + \frac{1}{2}(0.8)(20) + \frac{1}{4}(0.8)(10)$$

$$= 6 + 8 + 2$$

$$= 16$$